PS#10: Global optimization – Key

(a) $h(x) = xe^{-x}$ on the interval $[0, 3]$

(b) $p(t) = \sin(t) + \cos(t)$ on the interval $\left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$

(c) $q(x) = \dfrac{x^2}{x-2}$ on the interval $[3, 7]$

(d) $f(x) = 4 - e^{-(x-2)^2}$ with no domain restriction (so, no endpoints!)

(e) $h(x) = xe^{-ax}$ on the interval $\left[0, \dfrac{2}{a}\right]$, where $a > 0$ is some constant

(f) $f(x) = b - e^{-(x-a)^2}$ with no domain restriction, where a and b are some positive constants